MILITARY SPECIFICATION

ION EXCHANGE CARTRIDGE WITH FRAME ASSEMBLY FOR MERCURIC AND SILVER IONS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an ion exchange unit consisting of a cartridge frame assembly with one ion exchange cartridge containing resins capable of removing silver and mercuric ions from chloride-analysis waste boiler water.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116 - Preservation-Packaging, Methods of.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-S-1222 - Studs, Bolts, Hex Cap Screws, and Nuts.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of U.S. Navy.
MIL-STD-810 - Environmental Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following Government drawing forms a part of this specification to the extent specified herein.

DRAWING

MILITARY

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

577-4639 - Chloride-Analysis Waste Removal Assembly and Details.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN WELDING SOCIETY (AWS)

Welding Handbook

(Application for copies should be addressed to the American Welding Society, Inc., 250 N.W. 7th Street, Miami, FL 33125.)

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UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT
National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 1616 "P" Street, N.W., Washington, DC 20036.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish an ion exchange unit for first article inspection and approval (see 4.3 and 6.3).

3.2 Materials. Materials shall be as specified herein and on Drawing 577-4639.

3.2.1 Ion exchange cartridges. Ion exchange cartridges shall contain a layer of strong acid cation and a layer of intermediate base anion exchange resins in the hydrogen and hydroxyl cycles. Resins shall be unused, wet, provide maximum absorption and minimum leakage of all positive or negative ions to be removed or exchanged during the intended service.

3.2.2 Hardware. Hardware (bolts, nuts, screws, washers and miscellaneous hardware) shall be corrosion-resistant and of a good commercial grade material compatible with that of the basic fixture.

3.3 Design and construction.

3.3.1 Ion exchange cartridge unit. Ion exchange cartridge units shall conform to Drawing 577-4639 for design and dimensions. The cartridge unit shall consist of the following:

(a) A transparent plastic cartridge with a vane structure at both inlet and outlet ends. Both ends of the cartridge shall be capped, except for tapered nipples for adding and removing chloride-analysis waste water. Cartridges shall be filled with ion exchange resins as specified in 3.2.1. Unit content shall be as specified in 3.4.2.2.

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(b) A 60-mesh polyester screen shall be placed at the inlet and outlet ends of the cartridge on the inner side of the vane structure to retain the resin.
(c) Plastic funnels shall be placed at the inlet (top) of the cartridges to add chloride-analysis waste water to the ion exchange cartridges. The funnel mouth shall have a tab to secure it to the frame.
(d) Rubber tubing shall be connected to the outlet cap nipple of the cartridge in order to remove treated chloride-analysis waste water and to maintain the cartridge contents in a moist condition. When not in use, the open end of the tubing shall be secured near the top of the cartridge with the hose clip.
(e) A nameplate, "Chloride-Analysis Waste Removal Cartridge", fabricated from phenolic plastic, shall be affixed with epoxy cement on top center of the frame.

3.3.2 Cartridge frame assembly. Cartridge frame assembly shall conform to Drawing 577-4639 for design and dimensions. Cartridge frame assembly shall be bulkhead mounted.

3.3.2.1 Mounting bolts. Four 0.250 inch-20NC, 2-inch long, low carbon steel bolts in accordance with MIL-S-1222, shall be supplied for attachment of the frame to the bulkhead. Bolts shall not be welded or otherwise permanently affixed to the frame.

3.3.3 Dimensional tolerances. Unless otherwise specified (see 6.2.1), the following tolerances shall apply to dimensions specified on Drawing 577-4639:

(a) Fractional dimensions: plus or minus 1/64 inch.
(b) Decimal dimensions: plus or minus 0.005 inch.
(c) Angular dimensions: plus or minus 1/2 degree.

3.3.3.1 On Drawing 577-4639, a tolerance of plus or minus 1/32 inch is acceptable on fractional dimensions that are controlled by welding and brazing. Width tolerance shall not interfere with the interchangeability of assemblies or parts.

3.3.4 Sharp edges. Sharp edges and corners shall be given a slight radius.

3.3.5 Stress relief.

3.3.5.1 Metals. Metals used in fabrications and assemblies shall be treated to prevent deterioration of failure due to stresses or other conditions resulting from working, forming, or welding, during the fabrication process in accordance with AWS Welding Handbook.

3.3.5.2 Plastics. Processed plastics shall be stress-free to minimize deterioration of the parts and assembly.

3.3.6 Welding. Welding and allied processes used in fabrication shall be in accordance with MIL-STD-278, except that welder qualification shall not be needed.

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3.4 Performance.

3.4.1 Ion exchange unit.

3.4.1.1 Vibration. A complete ion exchange unit shall show no evidence of cartridge leakage, mechanical damage or loosening of structural parts following the test of 4.6.1.1.

3.4.1.2 Shock. A complete ion exchange unit shall pass the high shock tests for grade A, class 1 shockproofness specified in 4.6.1.2. Following the test, there shall be no signs of cartridge leakage, mechanical damage, breakage, or loosening of structural parts and distortion of sides.

3.4.2 Ion exchange cartridges.

3.4.2.1 Color. New (unused) cation resin shall be purple and turn yellow as its capacity is exhausted. New (unused) anion resin shall be yellow or brown and shall not change color in use (see 4.6.2.1).

3.4.2.2 Unit content. Unit content shall be 2,000 ± 50 mL, when tested as specified in 4.6.2.2.

3.4.2.3 Particle size. Not more than 5 percent of the resin shall pass a 50-mesh sieve and not more than 5 percent shall be retained on a 16-mesh sieve (see 4.6.2.3).

3.4.2.4 Moisture content. Moisture content for cation and anion exchange resins shall be as follows (see 4.6.2.4):

(a) Cation resin - 55 percent maximum, corresponding to a minimum solids content of 45 percent,
(b) Anion resin - 60 percent maximum, corresponding to a minimum solids content of 40 percent.

3.4.2.5 Whole bead content. Whole bead content shall be 70 percent, minimum (see 4.6.2.5).

3.4.2.6 Ion exchange cartridge performance capacity. When the ion exchange cartridge is tested for performance (see 4.6.2.6) a minimum of 55 grams of sodium chloride shall be removed from the 0.05 normal sodium chloride test solution before the specific resistivity of effluent is less than 20,000 ohms per centimeter (ohm/cm).

3.4.2.7 Shelf life. Ion exchange cartridges do not deteriorate significantly during storage, unless they lose the added water. In this event, the cartridges shall be discarded.

3.5 Designation and marking.

3.5.1 Identification plates. The complete ion exchange unit identification and information plate shall be as shown on Drawing 577-4639.

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3.6 Workmanship. Workmanship shall be in accordance with 3.6.1 through 3.6.3. Equipment shall comply with the visual and mechanical examination specified in 4.5.

3.6.1 Mounting of parts. Parts of hardware shall be assembled, secured, or mounted in the specified manner to accomplish the purpose for which intended. Equipment having missing, inoperative, defective, bent, broken or otherwise damaged parts shall not be acceptable.

3.6.2 Cleaning. After fabrication, parts and assembled equipment shall be cleaned of smudges, weld metal, metal chips, and any other foreign material which might detract from the intended operation, function, or appearance of the equipment.

3.6.3 Welding and brazing. Welds shall be free of harmful defects such as cracks, porosity, undercuts, voids, and gaps. There shall be no burn-through. Fillets shall be uniform and smooth. Angular or thickness misalinement, warpage, or dimensional change due to heat from the welding and brazing operation shall be within permitted tolerances. There shall be no damage to adjacent parts resulting from the welding or brazing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or puchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections shall be conducted in accordance with the test conditions specified in the General Requirements of MIL-STD-810.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in table I.

4.3.1 First article unit. First article unit shall consist of one ion exchange unit consisting of a frame asembly, and one ion exchange cartridge.

4.3.2 First article inspection report. The contractor shall furnish a first article inspection report in accordance with the data ordering document listed in the contract (see 6.2.2).

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TABLE I. Categories of inspection.

Category A (Complete ion exchange unit)	1st article inspection	Quality conformance	Requirement Paragraph	Test paragraph	Cartridge sample no.
Visual and mechanical examination	X	X	3.3, 3.5, 3.6	4.5	
Vibration	X		3.4.1.1	4.6.1.1	
Shock	X		3.4.1.2	4.6.1.2	
Category B (Ion exchange cartridges)					
Color	X	X	3.4.2.1	4.6.2.1	1
Unit content	X	X	3.4.2.2	4.6.2.2	1
Particle size	X	X	3.4.2.3	4.6.2.3	1
Moisture content	X	X	3.4.2.4	4.6.2.4	1
Whole bead content	X	X	3.4.2.5	4.6.2.5	1
Ion exchange cartridge performance capacity	X	X	3.4.2.6	4.6.2.6	2

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table I.

4.4.1 Lot. A lot shall consist of ion exchange units produced under essentially the same conditions, and offered for inspection at one time.

4.4.1.1 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4.2 Sampling for category A inspection. Sampling shall be in accordance with MIL-STD-105 for general inspection level I. The acceptable quality level (AQL) shall be 2.5 percent defective. Major and minor defects shall be as defined in MIL-STD-105.

4.4.3 Sampling for category B inspection. One sample shall be taken from each lot offered for delivery. A sample shall consist of four cartridges. Distribution of the cartridges shall be as follows:

(a) Two for inspection.
(b) One for the contractor to be held for shelf life test when required (see 4.6.2.7 and 6.2.1).
(c) One to be held by the contractor or "Government inspector" for 90 days to be used for retest in the case of disputes.

4.4.3.1 If a sample fails to pass any of the category B inspections, the sample shall be considered to have failed. Sample units which have been subjected to group B inspection shall not be delivered on the contract.

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4.4.4 Inspection of packaging. Sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. Sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Visual and mechanical examination. Each ion exchange unit shall be subjected to a thorough visual and mechanical examination to ascertain that the design and construction (see 3.3), designation and marking (see 3.5), and workmanship (see 3.6) conform to the requirements specified herein.

4.6 Test procedures.

4.6.1 Ion exchange unit.

4.6.1.1 Vibration. A complete ion exchange unit shall be tested in accordance with type I vibration of MIL-STD-167-1, up through 50 Hz. Performance requirements shall be as specified in 3.4.2.

4.6.1.2 Shock. A complete ion exchange unit shall be tested in accordance with the Class HI Shock Test for grade A, class 1, type C equipments specified in MIL-S-901. Performance requirements shall be as specified in 3.4.3.

4.6.2 Ion exchange cartridges. Ion exchange cartridges shall be tested as specified in 4.6.2.1 through 4.6.2.7. The following special test equipment or equal shall be used as required:

(a) Conductivity cell - Fisher Scientific Co. No. 9-324-26.
(b) Conductivity bridge - Fisher Scientific Co. No. 9-324-20.
(c) Filter - Whatman No. 41H filter paper.
(d) Sieves - U.S. Standard ASTM Sieves, type I, class 4, style A of RR-S-366.

4.6.2.1 Color. Resins in the cartridge shall be visually observed for color before and after the test of 4.6.2.6. Color of the ion exchange resin shall be as specified in 3.4.2.1.

4.6.2.2 Unit content. Unit content shall be determined as follows:

(a) Cut open a cartridge; keep the cation and anion exchange resins separated.
(b) Transfer into two 500 mL graduated cylinders, 500 + 5 mL of cation exchange resin and 500 + 5 mL of anion exchange resin. Store both resins in suitable containers to maintain resins in a moist condition.
(c) Add remaining resin into a 2000-mL graduated cylinder and measure volume.
(d) Calculate the unit content by summing the volumes of resin in the three graduated cylinders. Unit content shall be as specified in 3.4.2.2.

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4.6.2.3 Cation and anion exchange resins particle size. The test for particle size distribution of a mixed cation and anion exchange resin sample shall be performed as follows:

(a) Place in two graduate cylinders 50 ± 1 mL of cation and 50 ± 1 mL of anion exchange resins obtained in 4.6.2.2.
(b) Transfer the two resin samples to a 50-mesh sieve and place sieve in a shallow pan.
(c) Add tap water to the pan until the sieve is half submerged and the screen is covered with water.
(d) Raise and lower the sieve 20 times, just far enough to settle and float the particles of resin in the sieve.
(e) Remove the sieve and wash the contents of the pan into a beaker.
(f) Transfer the resin from the beaker into a 10-mL graduated cylinder and record the volume passing a 50-mesh sieve.
(g) Place a 16-mesh sieve in the pan and transfer to it the residue of resin left on the 50-mesh sieve.
(h) Half submerge the 16-mesh sieve, raise and lower until further action results in a negligible number of particles passing through the sieve.
(i) Invert the 16-mesh sieve in a clean pan and wash the retained particles into the pan.
(j) Decant the excess water from the pan and transfer the coarse resin particles to a beaker.
(k) Transfer the resin from the beaker into a 10-mL graduated cylinder and record the volume passing a 16-mesh sieve.
(l) Calculate the particle size percentages as follows:

The volume of resin in mL passing 50-mesh sieve = percent resin passing 50-mesh sieve. The volume of resin in mL retained on 16-mesh sieve = percent resin on 16-mesh sieve. The percentage shall be as specified in 3.4.2.3.

4.6.2.4 Moisture content. The test for moisture content of cation and anion exchange resins shall be performed as follows:

(a) Weigh 50.0 ± 0.1 gram of resin obtained in 4.6.2.2.
(b) Place the resin on a large fritted glass funnel with a coarse filter plate or a 60-degree funnel with a filter paper, and wash with several portions of acetone to remove the water.
(c) Air-dry until the sample no longer gives off the odor of acetone and transfer to a tared weighing bottle.
(d) Dry in an oven at 221°F for 16 hours, cool in a desiccator and reweigh.
(e) Calculate percent solids as follows:

$$\text{Percent moisture} = \frac{(50 \text{ grams} - \text{dry weight of resin sample in grams}) \times 100}{50 \text{ grams}}$$

Percent moisture shall be as specified in 3.4.2.4.

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4.6.2.5 Whole bead content. Whole bead content for the ion exchange resin shall be determined as follows:

(a) Position a highly polished metal plate, at least 1.5 feet long by 1 foot wide, such that the surface makes a 10 degree angle with the horizontal.
(b) Weigh into separate 50-mL beakers 10 ± 0.1 grams of cation and anion exchange resins that have been allowed to dry in air until the particles have no tendency to adhere to one another. Combine the two resins in one 50-mL beaker.
(c) Pour the resin sample onto the top of the inclined plane slowly so that the beads flow down the plane without interfering with each other.
(d) Collect the beads that roll off the plate and obtain total weight to nearest 0.1 gram.
(e) Calculate the whole bead content as follows:

$$\text{Percent of whole beads} = \frac{\text{weight of beads rolling off plate} \times 100}{20 \text{ grams}}$$

Whole bead content shall be as specified in 3.4.2.5.

4.6.2.6 Ion exchange cartridge performance capacity. The available ion exchange capacity of a cartridge shall be determined as follows:

(a) Remove plastic caps used for moisture retention from the hose connections of the cartridge, and mount the cartridge in a vertical position.
(b) Prepare the 0.05 normal sodium chloride test solution by dissolving 87.68 grams of dried chemically pure sodium chloride in 30 liters (L) of distilled or deionized water.
(c) Place this solution in a suitable container with draw-off connections.
(d) Mount the container about two feet above the top of the cartridge.
(e) Make the hose connections necessary to deliver the 0.05 normal sodium chloride solution to the top of the cartridge, incorporating a screw clamp or other valve to regulate flow.
(f) Fit a conductivity cell through a three-hole rubber stopper inserted into a small flask or bottle. Make glass and rubber tubing connections to permit flow of effluent from the cartridge to the bottom of the flask or bottle and out the top into a receiver of 30-L capacity.
(g) Connect the conductivity cell to a conductivity bridge. Adjust the flow of the 0.05 normal sodium chloride solution through the cartridge to 60 mL per minute (mL/min), collecting the effluent in the receiver.
(h) Make periodic readings of the resistivity of the effluent, decreasing the interval between readings to about 1 per minute as the volume of effluent approaches the 19-L mark.
(i) When the conductivity bridge indicates a specific resistivity of 20,000 ohms/cm for the effluent, stop the flow of 0.05 normal sodium chloride solution and measure the volume of the effluent which has passed through the cartridge.

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(j) Calculate the amount of sodium chloride removed as follows:

Ion exchange cartridge performance capacity (grams of sodium chloride removed) = liters of effluent water x 2.923. The available ion exchange capacity per cartridge shall be as specified by 3.4.2.6.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging. Preservation-packaging shall be level A, or C, as specified (see 6.2.1).

5.1.1 Level A. Ion exchange cartridge units shall be individually packaged in accordance with method III of MIL-P-116 as follows:

Each unit shall be placed in an end opening type CF, class domestic, style FOL fiberboard box conforming to PPP-B-636. Attaching hardware shall be packaged in a transparent plastic bag, minimum 4 mil thick and secured to the cartridge frame. Box closure shall conform to method I as specified in the appendix to the box specification, except that staples shall not be used.

5.1.2 Level C. Ion exchange cartridge units shall be individually packaged to afford protection against loss of contents and physical damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal retail or wholesale packaging methods may be utilized when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A. Ion exchange cartridge units packaged as specified in 5.1 shall be packed in containers conforming to any one of the following specifications at the option of the contractor.

Specification	Container	Type or class
PPP-B-601	Wood, cleated plywood	Overseas type
PPP-B-621	Wood, nailed and lock corner	Class 2
PPP-B-636	Fiberboard	Weather-resistant
PPP-B-640	Fiberboard-corrugated triplewall	Class 2

5.2.1.1 Weight. Maximum gross weight, container plus contents, shall not exceed 200 pounds. For fiberboard containers, the maximum gross weight shall not exceed the applicable requirements for the container grade selected.

5.2.1.2 Closure and case liners. Shipping containers shall be closed, strapped, banded, or reinforced in accordance with the applicable container specification or appendix thereto with method V closure

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applicable to PPP-B-636 containers. Unless otherwise specified (see 6.2.1), shipping containers shall have case liners conforming to and closed and sealed in accordance with MIL-L-10547. Case liners for fiberboard containers may be omitted provided all center and edge seams and manufacturer's joints are sealed as specified for method V in the appendix to PPP-B-636.

5.2.2 Level B. Ion exchange units packaged as specified in 5.1 shall be packed in containers as specified for level A, except that the containers shall be of the domestic class or grade and case liners strapping, waterproofing and reinforcing is not required. Container closure shall be in accordance with the applicable container specification or appendix thereto with method I closure applicable to PPP-B-636 containers.

5.2.3 Level C. Ion exchange units, packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Cushioning and wrapping materials. For items destined for shipboard stowage or installation, use of excelsior, newspaper, shredded paper (all types including wax paper) and similar hygroscopic or nonneutral materials and all types of loose-fill materials, for applications such as cushioning, stuffing, and dunnage is prohibited. Materials selected for cushioning and wrapping shall have properties (characteristics) resistant to fire. When loose-fill material is permitted (see 6.2.1), containers shall be marked or labelled as follows:

"CAUTION

Contents cushioned with "loose-fill" material. Loose-fill not to be taken aboard ship. Remove and discard cushioning. If required, recushion with cellulosic material, bound fiber, transparent flexible cellular material, and so forth."

5.4 Marking. In addition to any special marking required (see 6.2.1) and herein, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.4.1 Special marking.

(a) Unit packages and exterior shipping container markings shall include the following:
    (1) *Batch number*
    (2) Date of manufacture
    (3) Net contents
(b) Each unit container (see 5.1.1) shall have the closure end of the container marked, "OPEN THIS END".

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6. NOTES

6.1 Intended use. Ion exchange units are used to remove mercuric and silver ions from chloride-analysis wastewater.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of specification.
(b) When a first article is required for inspection and approval (see 3.1, 4.3.1, and 6.3).
(c) If tolerances apply to other than dimensions specified (see 3.3.3).
(d) If inspection conditions are to be other than specified (see 4.1.1).
(e) Level of preservation-packaging required (see 5.1).
(f) Level of packing required (see 5.2).
(g) When case liners are not required (see 5.2.1.2).
(h) When loose-fill material is permitted (see 5.3).
(i) Special marking, if required (see 5.4).

6.2.2 When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph	Data requirement	Applicable DID	Option
4.3	First article inspection report	DI-T-4902	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection, as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products who wish to rely on such productions or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending acquisition.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 4610-N074)

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